复分析

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9.11 导论 方汉隆

内容 分 · Beltrami equation
· Bers simultaneously uniformization
析 · Teichmüler theory
几何 · Hullbard's Vol I, III Techimüler Theory

教材 Elliptic Partial Differential Equations and Quasiconformal Mapping in the plane
Kari Astala ; Tadeusz Iwaniec ; Gaven Martin

评分标准 研究生 90 报告 95-100
本科生 85 报告 95-100 (1.5~2h)

报告内容 ① 教材未讲部分 ② Bers某工作 ③ Hullbard某部分

参考书 Lipman Bers: a life in Mathematics Keen. Kra. Rodriguez

邮箱 hanlongfang@math.pku.edu.cn 智华楼441 周五下午3:30-4:30
助教 2501110031@stu.pku.edu.cn

1. ① h在$\Omega\subset\mathbb{C}$全纯 $\Leftrightarrow$ 满足齐次C-R方程 $\frac{\partial h}{\partial \bar{z}}=\frac{1}{2}\left(\frac{\partial h}{\partial x}+i\frac{\partial h}{\partial y}\right)=0$.

Beltrami方程 $\frac{\partial h}{\partial \bar{z}}=\mu\frac{\partial h}{\partial z}$

② 非齐次C-R方程 $\frac{\partial u}{\partial \bar{z}}=f$ f连. (或 $\partial u=f\,d\bar{z}$)

③ Cauchy核 $f(z)=\frac{1}{z-a}$. 在$\mathbb{C}\setminus\{a\}$全纯. a为一阶极点.

④ 定义: 称f局部可积 若对任意紧集$K\subset\Omega$. $\iint_K |f|\,dx\,dy<+\infty$

记f为 $L^1_{loc}(\Omega)$

2. ① 引理: $\forall R>0$ $a\in\mathbb{C}$ $I_R=\iint_{B_R(a)}\frac{1}{|z-a|}\,dx\,dy=2\pi R$.

故 $\frac{1}{z-a}$ 局部可积 但不可积

证明: $z=a+re^{i\theta}$ $I_R=\int_0^{2\pi}d\theta\int_0^R\frac{1}{r}\,r\,dr=2\pi R$

② 引理: Cauchy核的微分 $\frac{\partial}{\partial z}\left(\frac{1}{z-a}\right)=-\frac{1}{(z-a)^2}$ 不局部可积

($\int_{B_R(a)}\frac{1}{|z-a|^2}\,dx\,dy=\int_0^{2\pi}d\theta\int_0^R\frac{1}{r^2}\,r\,dr=2\pi\int_0^R\frac{dr}{r}$ 发散)

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③ 引理：$f\in C_c(\mathbb{C})$ 令 $u(z)=\iint_{\mathbb{C}}\frac{f(\xi)}{\xi-z}d\xi\wedge d\bar{\xi}$ $z\in\mathbb{C}$

则 u 有界，满足 $\sup_{z\in\mathbb{C}}|u(z)|\le C\sup_{z\in\mathbb{C}}|f(z)|$

其中 C 连续依赖于 f 支集的直径，$u(z)$ 在 $\mathbb{C}$ 连续

证明：设 $\operatorname{supp} f\subset B_R(0)$ $\sup_{z\in\mathbb{C}}|f(z)|\le M$

$$|u(z)|\le 2\iint_{\{\xi:|z|-R\le|\xi-z|\le|z|+R\}}\frac{M}{|\xi-z|}dx\,dy=2\int_0^{2\pi}d\theta\int_{|z|-R}^{|z|+R}M\,dr=(8\pi R)M \text{ 有界}$$

下证 u 连续，取定 z_0 由 $u(z)\xrightarrow{\xi=z+\eta}\iint_{\mathbb{C}}f(z+\eta)/\eta\,d\eta\wedge d\bar{\eta}$

设 $\{z_n\}$ 为收敛至 z_0 的点列，取 n 充分大使

$\operatorname{supp} f(z_n+\eta)/\eta\subset B_{|z_0|+1+R}(0)$ 从而 $\left|\frac{f(z_n+\eta)}{\eta}\right|\le\frac{M}{|\eta|}1_{B_{|z_0|+1+R}(0)}$

所以 $\left|\iint_{\mathbb{C}}\frac{f(\xi)}{\xi-z_n}d\xi\wedge d\bar{\xi}\right|\le 2\iint_{\mathbb{C}}\frac{M}{|\eta|}1_{B_{|z_0|+1+R}(0)}|d\eta\wedge d\bar{\eta}|<+\infty$

由控制收敛定理 $u(z_n)\to u(z_0)$ $n\to+\infty$. u 连续

3. ① 称 $u\in C^k(\Omega)$ $k\in\mathbb{N}\cup\{\infty\}$ 若 u 在 Ω 邻域为 C^k 函数

② 定理（Cauchy 积分公式） $f(z_0)=\frac{1}{2\pi i}\int_\Gamma\frac{f(z)}{z-z_0}dz$ $\Gamma=\{z:|z-z_0|=\rho\}$

③ 定理（Cauchy-Poincaré 公式） Ω 为 $\mathbb{C}$ 上有界开集，$\partial\Omega$ 分段 C^1 $u\in C^1(\overline{\Omega})$

$$u(z)=\frac{1}{2\pi i}\left(\int_{\partial\Omega}\frac{u(\xi)}{\xi-z}d\xi+\iint_\Omega\frac{\partial u(\xi)/\partial\bar{\xi}}{\xi-z}d\xi\wedge d\bar{\xi}\right).$$

证明：取定 z，令 $\Omega_\varepsilon=\Omega\setminus\overline{B_\varepsilon(z)}$ $0<\varepsilon\ll1$ 由 Stokes $\int_{\partial\Omega_\varepsilon}\phi=\iint_{\Omega_\varepsilon}d\phi$

其中 $\phi=\frac{u(\xi)}{\xi-z}d\xi$ $d\phi=\frac{\partial u}{\partial\bar{\xi}}\frac{d\bar{\xi}\wedge d\xi}{\xi-z}$ 代入得

$$\int_{\partial\Omega}\frac{u(\xi)}{\xi-z}d\xi-\int_{\partial B_\varepsilon}\frac{u(\xi)}{\xi-z}d\xi=\iint_{\Omega_\varepsilon}\frac{\partial u}{\partial\bar{\xi}}\frac{d\bar{\xi}\wedge d\xi}{\xi-z}=\left(\iint_\Omega-\iint_{B_\varepsilon(0)}\right)\frac{\partial u}{\partial\bar{\xi}}\frac{d\bar{\xi}\wedge d\xi}{\xi-z}$$

- $\lim_{\varepsilon\to0}\int_{\partial B_\varepsilon(0)}\frac{u(\xi)}{\xi-z}d\xi=\lim_{\varepsilon\to0}\int_0^{2\pi}\frac{u(z+\varepsilon e^{i\theta})}{\varepsilon e^{i\theta}}\varepsilon i e^{i\theta}d\theta=2\pi i\,u(z)$

- $\lim_{\varepsilon\to0}\iint_{B_\varepsilon(0)}\frac{\partial u}{\partial\bar{\xi}}\frac{d\bar{\xi}\wedge d\xi}{\xi-z}=\lim_{\varepsilon\to0}\int_0^{2\pi}\int_0^{\varepsilon}\frac{\partial u}{\partial\bar{\xi}}\cdot\frac{-2\rho i e^{2i\theta}d\rho d\theta}{\rho e^{i\theta}}=0.$

□

9.16 全纯延拓 对数势

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1. ① 设 $C_0^k(\Omega)$ 为 Ω 上紧支 k 阶光滑函数

② 定理：$f\in C_0^k(\mathbb{C})$ $k\geq 1$. 令 $u(z)=\frac{1}{2\pi i}\iint_{\mathbb{C}}\frac{f(\zeta)}{\zeta-z}d\zeta\wedge d\bar{\zeta}$ 则 $\frac{\partial u}{\partial \bar{z}}=f$

证明：取 R 充分大 $\operatorname{supp} u\subset B_R(a)=\{z:|z-a|<R\}\subset\mathbb{C}$

$$\frac{\partial u}{\partial \bar{z}}(z)=\frac{\partial}{\partial \bar{z}}\left(\frac{1}{2\pi i}\iint_{\mathbb{C}}\frac{f(z+\eta)}{\eta}d\eta\wedge d\bar{\eta}\right)=\frac{1}{2\pi i}\iint_{\mathbb{C}}\frac{\partial f(z+\eta)}{\partial \bar{z}}\frac{d\eta\wedge d\bar{\eta}}{\eta}=\frac{1}{2\pi i}\iint_{\mathbb{C}}\frac{\partial f(\zeta)}{\partial \bar{\zeta}}\frac{d\zeta\wedge d\bar{\zeta}}{\zeta-z}$$

$$=\frac{1}{2\pi i}\iint_{B_R(a)}\frac{\partial f(\zeta)}{\partial \bar{\zeta}}\frac{d\zeta\wedge d\bar{\zeta}}{\zeta-z}=f(z)-\frac{1}{2\pi i}\int_{\partial B_R(a)}\frac{f(\zeta)}{\zeta-z}d\zeta=f(z).$$

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③ 推论：$f\in C_0^\infty(\mathbb{C})$ $\exists u\in C^\infty(\mathbb{C})$ $\frac{\partial u}{\partial \bar{z}}=f$ 有

2. ① 有界单连通区域 $\Omega\subset\mathbb{C}$. $\partial\Omega$ C^1，设 f 为 $\partial\Omega$ 上连续函数 定义 Cauchy 变换

$$F(z)=\frac{1}{2\pi i}\int_{\partial\Omega}\frac{f(\zeta)}{\zeta-z}d\zeta \qquad z\notin\partial\Omega$$

$\Omega^+=\bar{\Omega}^c$ $\Omega^-=\Omega$ 记 F^-, F^+ 为 F 在 Ω^- Ω^+ 上限制

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② 引理：F^-, F^+ 均全纯 $\lim_{z\to\infty}F^+(z)=0$.

证明：设 $\bar{\Omega}\subset B_R(0)$ $\sup_{\zeta\in\partial\Omega}|f(\zeta)|\leq M$ $\forall|z|>2R$.

$$|F^+(z)|=|F(z)|\leq\frac{1}{2\pi}\int_{\partial\Omega}M\frac{|d\zeta|}{|z|-R}\to 0.$$



③ 定理(Jump公式) 设 F^+ 与 F^- 可连续延拓至边界 且

$$f(z)=F^-(z)-F^+(z)$$

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证明：· 若 f 全纯于 $\bar{\Omega}$，$F^-=f$. $F^+\equiv 0$. 成立.

· 设 $f_e(z)$ 为 f 在 $\mathbb{C}$ 的延拓. 令 $\Phi(z)=\frac{1}{2\pi i}\int_{\partial\Omega}\frac{f(\zeta)-f_e(\zeta)}{\zeta-z}d\zeta$.

Φ 在 $\mathbb{C}\setminus\partial\Omega$ 连续 对 $z\in\partial\Omega$. 由 $|f(\zeta)-f_e(z)|\leq M|\zeta-z|$

由控制收敛定理 Φ 连续.

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$$F^-(z)=\frac{1}{2\pi i}\int_{\partial\Omega}\frac{f(\zeta)-f_e(\zeta)}{\zeta-z}d\zeta+f_e(z)=\Phi(z)+f_e(z)$$

$F^+(z)=\Phi(z)$. 取极限即证.

④ 定理：以下三条等价 条件同①. Ω 有界单连通 $\partial\Omega$ C^1 $f\in C^1(\partial\Omega)$

(i) f 为 $F\in C(\bar{\Omega})\cap O(\Omega)$ 在 $\partial\Omega$ 限制

(ii) $\int_{\partial\Omega}f(z)z^n dz=0$ (iii) $f=F^-|_{\partial\Omega}$

证明：(i)⇒(ii) ✓ (iii)⇒(i)✓

(ii)⇒(iii). 设 $\Omega \subseteq B_{\frac{R}{2}}(0)$. 则 $\forall |z| \geq R \quad \left|\frac{\xi}{z}\right| \leq \frac{1}{2}$.

$$F^+(z) = \frac{1}{2\pi i}\int_{\partial\Omega} \frac{f(\xi)}{\xi - z} d\xi = \frac{1}{2\pi i}\int_{\partial\Omega} \frac{1}{z}\,\frac{f(\xi)}{1-\frac{\xi}{z}} d\xi = -\frac{1}{2\pi i}\int_{\partial\Omega} \sum_{n=0}^{\infty} f(\xi)\xi^n z^{-n-1} d\xi$$

$$= -\frac{1}{2\pi i}\sum_{n=0}^{\infty}\left(\int_{\partial\Omega} f(\xi)\xi^n d\xi\right) z^{-n-1} = 0 \quad \Rightarrow f(z) = F^-(z) - F^+(z) = F^-(z)$$

⑤ 定理：$f \in C_0^k(\mathbb{C})$ $\exists u \in C_0^k(\mathbb{C})$ 使 $\frac{\partial u}{\partial \bar{z}} = f$ 当且仅当 $\iint_{\mathbb{C}} f(z) z^n dz \wedge d\bar{z} = 0 \ \forall n$

证明：⇒. $f(z) z^n dz \wedge d\bar{z} = \frac{\partial u}{\partial \bar{z}} z^n dz \wedge d\bar{z} = d(u z^n dz)$ 又 $u z^n dz$ 紧支. 积分。

⇐ 设 $\operatorname{supp} f \subset B_{\frac{R}{2}}(0)$. $\forall |z| > R$. $\forall \xi \in \operatorname{supp} f$. $\left|\frac{\xi}{z}\right| < \frac{1}{2}$

$$u(z) = \frac{1}{2\pi i}\iint_{\mathbb{C}} \frac{f(\xi)\, d\xi \wedge d\bar{\xi}}{\xi - z} = -\frac{1}{2\pi i}\iint_{\operatorname{supp} f} \frac{f(\xi)}{1-\frac{\xi}{z}}\,\frac{1}{z}\, d\xi \wedge d\bar{\xi}$$

$$= -\frac{1}{2\pi i}\iint_{\operatorname{supp} f} f(\xi)\sum_{n=0}^{\infty} \xi^n z^{-n-1} d\xi \wedge d\bar{\xi} = -\frac{1}{2\pi i}\sum_{n=0}^{\infty}\iint_{\mathbb{C}} f(\xi)\xi^n d\xi \wedge d\bar{\xi} \cdot z^{-n-1} = 0$$

故 $\operatorname{supp} u \subset B_R(0)$ 证毕.

3. ① 定义（对数势） 给定 $\phi \in C_0^\infty(\mathbb{C})$ 定义对数势

$$(\mathcal{L}\phi)(z) = \frac{2}{\pi}\int_{\mathbb{C}} \phi(z) \log|z - z|\, dA(z) \quad dA(z) = dx\,dy$$

注：$\mathcal{L}\phi$ 良定 (令 $z = z + \rho e^{i\theta}$ 即可)

② 引理 $\mathcal{L}(D\phi) = D(\mathcal{L}\phi)$ $\phi \in C_0^\infty(\mathbb{C})$ $D = \sum_{0 \leq k,l < \infty} a_{k,l} \frac{\partial^{k+l}}{\partial z^k \partial \bar{z}^l}$

证明：记 $(T_w\phi)(z) = \phi(z + w)$

$$\mathcal{L}(T_w\phi)(z) = \frac{2}{\pi}\int_{\mathbb{C}} \phi(w + z) \log|z - z|\, dA(z) = \frac{2}{\pi}\int_{\mathbb{C}} \phi(z) \log|z + w - z|\, dA(z) = T_w(\mathcal{L}\phi)(z)$$

对 $w = tx$ ty 求导即可

③ 引理：$\mathcal{L}(\phi) \in C^\infty(\mathbb{C})$ $\forall \phi \in C_0^\infty(\mathbb{C})$

证明：以 $D = \frac{\partial}{\partial x}$ 为例. $D(\mathcal{L}\phi)(z) = \lim_{h \to 0} \frac{\mathcal{L}\phi(z+h) - \mathcal{L}\phi(z)}{h} = \varphi_h(z)$

$$|\varphi_h(z)| \leq C \sup_{\hat{z} + z \in \operatorname{supp}\phi = K} \left|\frac{\partial}{\partial x}\phi(\hat{z} + z)\right| \int |\log|z||\, dA(z) < +\infty.$$ 控制收敛定理知连续.

④ 引理：$\frac{\partial^2}{\partial z \partial \bar{z}} \mathcal{L}\phi = \phi$.

9.18 Cauchy 变换. Beurling 变换

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1. ① Green 对数势 $(L\phi)(z)=\frac{2}{\pi}\int_{\mathbb{C}}\phi(\zeta)\log|z-\zeta|\,dA(\zeta)$ $\phi\in C_0^\infty(\mathbb{C})$

② 引理: $\frac{\partial^2}{\partial z\partial\bar{z}}L\phi=\phi$

证明: $LHS=L\left(\frac{\partial^2}{\partial z\partial\bar{z}}\phi\right)=\frac{1}{2\pi}\int_{\mathbb{C}}\Delta\phi(\zeta)\log|z-\zeta|\,dA(\zeta)$ 设 $\operatorname{supp}\phi\subset B_R(0)$

$\Omega_\varepsilon=B_R(0)\setminus\overline{B_\varepsilon(z)}$ 由 $\Delta\log|z-\zeta|=0$

$\nabla\cdot(\nabla\phi(\zeta)\log|z-\zeta|)=\Delta\phi\log|z-\zeta|+\nabla\phi\cdot\nabla\log|z-\zeta|=\Delta\phi\log|z-\zeta|+\nabla\cdot(\nabla\log|z-\zeta|\phi)$

Green 公式

$$\frac{1}{2\pi}\int_{\Omega_\varepsilon}\Delta\phi(\zeta)\log|z-\zeta|\,dA(\zeta)=\frac{1}{2\pi}\int_{\partial\Omega_\varepsilon}\log|z-\zeta|\,\nabla\phi\cdot\vec{n}\,ds-\frac{1}{2\pi}\int_{\partial\Omega_\varepsilon}\nabla\log|z-\zeta|\cdot\vec{n}\phi\,ds$$

$\varepsilon\to0$ 得 $LHS=\frac{1}{2\pi}\lim_{\varepsilon\to0}\int_{\Omega_\varepsilon}\Delta\phi(\zeta)\log|z-\zeta|\,dA(\zeta)=-\lim_{\varepsilon\to0}\frac{1}{2\pi}\int_{\partial\Omega_\varepsilon}\frac{\phi(\zeta)}{|z-\zeta|}\,ds=\phi(z)$

③ 引理: $L\phi(z)=\left|\frac{2}{\pi}\int_{\mathbb{C}}\phi(\zeta)\,dA(\zeta)\right|\log|z|+O(1)$ $z\to\infty$

证明: $L\phi(z)-\frac{2}{\pi}\int_{\mathbb{C}}\phi(\zeta)dA(\zeta)\log|z|=\frac{2}{\pi}\int_{\mathbb{C}}\phi(\zeta)\log\left|1-\frac{\zeta}{z}\right|dA(\zeta)\to0$.

推论: $L\phi$ 在 $\mathbb{C}$ 一致连续.

2. ① 定义 Cauchy 变换 $C\phi(z)=\frac{\partial L\phi}{\partial z}=\frac{1}{\pi}\int_{\mathbb{C}}\frac{\phi(\zeta)}{z-\zeta}\,dA(\zeta)$.

注: $\frac{\partial L\phi}{\partial z}=L(\phi_z)=\frac{1}{\pi}\int_{\mathbb{C}}\phi_z(\zeta)\log|z-\zeta|^2dA(\zeta)=\frac{1}{\pi}\lim_{\varepsilon\to0}\int_{\Omega_\varepsilon}\phi_z(\zeta)\log|z-\zeta|^2dA(\zeta)$

$$=\lim_{\varepsilon\to0}-\frac{1}{\pi}\int_{\partial\Omega_\varepsilon}\phi(\zeta)\frac{\partial}{\partial\zeta}\log|z-\zeta|^2dA(\zeta)+\lim_{\varepsilon\to0}\int_{\partial\Omega_\varepsilon}\underbrace{\phi(\zeta)}_{(\to0)}\log|z-\zeta|^2ds$$

$$=\frac{1}{\pi}\int_{\mathbb{C}}\frac{\phi(\zeta)}{z-\zeta}\,dA(\zeta)$$

② 引理: $C(D\phi)=D(C\phi)$ $D=\sum a_{k,l}\frac{\partial^{k+l}}{\partial z^k\partial\bar{z}^l}$

证明: $C(D\phi)=L\left(\frac{\partial}{\partial z}D\phi\right)=D\left(\frac{\partial}{\partial z}L\phi\right)=D(C\phi)$

③ 在 $C_0^\infty(\mathbb{C})$ 定义内积 $\langle\phi,\eta\rangle=\int_{\mathbb{C}}\bar{\eta}\phi\,dA(\zeta)$

定义 C 的对偶 $\langle C\phi,\eta\rangle=\langle\phi,C^*\phi\rangle$

④ 性质: $C^*\eta=\overline{C\bar{\eta}}=\frac{1}{\pi}\int_{\mathbb{C}}\frac{\eta(\zeta)}{\bar{z}-\bar{\zeta}}dA(\zeta)$

证明: $\int_{\mathbb{C}}\bar{\eta}C\phi\,dA(\zeta)=\int_{\mathbb{C}}\bar{\eta}\frac{\partial L\phi}{\partial z}dA(\zeta)=-\int_{\mathbb{C}}\frac{\partial}{\partial z}\bar{\eta}L\phi\,dA(\zeta)=-\int_{\mathbb{C}}\overline{\frac{\partial\eta}{\partial\bar{z}}}L\phi\,dA(\zeta)$

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而 $\int_{\mathbb{C}} \chi(z)(\mathcal{I}\phi)(z)\,dA(z) = \int_{\mathbb{C}} \chi(z)\int_{\mathbb{C}} \phi(\zeta)\log|\zeta-z|\,dA(\zeta)\,dA(z)$

$\xrightarrow{\substack{\zeta-z=u\\ z+\zeta=w}}$ $\int_{\mathbb{C}\times\mathbb{C}} \chi\left(\frac{u+w}{2}\right)\phi\left(\frac{w-u}{2}\right)\log|u|\,dA(u)\,dA(w)$

Fubini: $\int_{\mathbb{C}}\phi(\zeta)\int_{\mathbb{C}}\chi(z)\log|z-\zeta|\,dA(z)\,dA(\zeta) = \int_{\mathbb{C}}\phi(\zeta)\mathcal{I}(\chi)\,dA(\zeta)$

故 $\int_{\mathbb{C}}\overline{\eta}\,\mathcal{C}\phi\,dA(z) = -\int\mathcal{I}(\overline{\eta_z})\phi\,dA = -\int_{\mathbb{C}}\mathcal{C}\overline{\eta}\,\phi\,dA(z) \Rightarrow \mathcal{C}^*\eta = \overline{\mathcal{C}\overline{\eta}}$

$\mathcal{C}^*\eta = \overline{\mathcal{C}\overline{\eta}} = \frac{1}{\pi}\int_{\mathbb{C}}\frac{\overline{\phi(z)}\,\eta(z)}{z-\zeta}\,dA(z) = \frac{1}{\pi}\int_{\mathbb{C}}\frac{\eta(z)}{\overline{z}-\overline{\zeta}}\,dA(z)$

3. ① 定义 Beurling 变换 $(S\phi)(z) = \frac{\partial^2}{\partial z^2}\mathcal{I}\phi = \frac{\partial}{\partial z}\mathcal{C}\phi = -\frac{1}{\pi}\int_{\mathbb{C}}\frac{\phi(\zeta)}{(z-\zeta)^2}dA(\zeta)$

但核奇异可积，取 Cauchy 主值

$(S\phi)(z) = \mathcal{C}(\phi_{\bar z})(z) = \lim_{\varepsilon\to 0}\frac{1}{\pi}\int_{\mathbb{C}\setminus D(z,\varepsilon)}\frac{\phi_{\bar z}(\zeta)}{z-\zeta}\,dA(\zeta) = -\frac{1}{\pi}\lim_{\varepsilon\to 0}\int_{|z-\zeta|>\varepsilon}\frac{\phi(\zeta)}{|z-\zeta|^2}\,dA(\zeta)$

② 引理：$(S\circ\frac{\partial}{\partial z})\phi = \frac{\partial}{\partial z}\phi$

证明：$\because S\phi_{\bar z} = \mathcal{C}\phi_{\bar z z} = \frac{\partial}{\partial z}\mathcal{I}\phi_{z\bar z} = \frac{\partial^2}{\partial z\partial\bar z}\mathcal{I}\phi_z = \phi_z$.

③ 引理：S 在内积 $B(f,g) = \int_{\mathbb{C}} f\overline{g}\,dA(z)$ 下对称

证明：$\int_{\mathbb{C}}\eta\,S\phi\,dA(z) = -\frac{1}{\pi}\int_{\mathbb{C}}\int_{|z-\zeta|>\varepsilon}\frac{\eta(z)\,\phi(\zeta)}{(z-\zeta)^2}\,d\zeta\,dz = \int_{\mathbb{C}} S\eta\,\phi\,dA(\zeta)$.

④ 命题：$S: L^2(\mathbb{C}) \to L^2(\mathbb{C})$ 为同构 $\|S\phi\|_{L^2} = \|\phi\|_{L^2}$ (*)

证明：$\int_{\mathbb{C}}|f_z|^2\,dA(z) = \int_{\mathbb{C}} f_z\overline{f}_{\bar z}\,dA(z) = \int_{\mathbb{C}} f_{\bar z}(\overline{f})_z\,dA(z) = \int_{\mathbb{C}}|f_{\bar z}|^2\,dA(z)$

又 $S\phi_z = \phi_{\bar z}$ 知 $\|S\phi_z\|_{L^2} = \|\phi_{\bar z}\|_{L^2} = \|\phi_z\|_{L^2}$ 从而 (*) 在 $R = \{\partial f : f\in C_0^\infty(\mathbb{C})\}$ 对下证 R 在 $L^2(\mathbb{C})$ 中稠密. 若 $h\in L^2(\mathbb{C})$ 与全体 R 中元素正交，$h_{\bar z}$ 分布意义下为 0. 由 Weyl 引理，h 全纯. $h\in L^2(\mathbb{C})\cap G(\mathbb{C})$

平均值原理 $f(0) = \frac{1}{B(0,r)}\int_{B(0,r)} f(z)\,dA(z) \le \frac{1}{B(0,r)^{\frac12}}\sqrt{\int_{B(0,r)}|f|^2\,dA(z)} \to 0$.

故 $L^2(\mathbb{C})\cap G(\mathbb{C}) = \{0\}$ $h = 0$.

⑤. 由因 S 可延拓为 $L^2(\mathbb{C})$ 上连续满算子. S 为酉算子

伴随 S^*. 逆 S^{-1}. 我们证明 $S^{-1}=S^*=\frac{1}{\pi}\int_{\mathbb{C}}\frac{\partial\phi(z)}{(\bar z-\bar z)^2}dz=\overline{S\phi}(z)$

证明: $\langle S\phi,\psi\rangle=\int_{\mathbb{C}}\lim_{\varepsilon\to0^+}\frac{1}{\pi}\int_{|z-z|>\varepsilon}\frac{\phi(z)}{(z-z)^2}dA(z)\overline{\psi(z)}\,dA(z)$

$=\int_{\mathbb{C}}\phi(z)\lim_{\varepsilon\to0^+}\frac{1}{\pi}\int_{|z-z|>\varepsilon}\frac{\overline{\psi(z)}}{(z-z)^2}dA(z)\,dA(z)\Rightarrow S^*\psi=\lim_{\varepsilon\to0^+}\frac{1}{\pi}\int_{|z-z|>\varepsilon}\frac{\overline{\psi(z)}}{(z-z)^2}dA(z)$

$\langle S\phi,S\psi\rangle=\langle\phi,\psi\rangle=\langle\phi,SS^*\psi\rangle\Rightarrow SS^*=I$

$\Rightarrow S^{-1}=S^*=\lim_{\varepsilon\to0}\frac{1}{\pi}\int_{|z-z|>\varepsilon}\frac{\psi(z)}{(\bar z-\bar z)^2}dA(z)$.

⑥. 注1. $R=\{\bar\partial f:f\in C_0^\infty(\mathbb{C})\}$ 事实上在 $L^q(\mathbb{C})$ $1<q<+\infty$ 稠密. $(\Delta|f|^p\ge0\ p\ge2)$

注2. $p=\infty$. $1\in L^\infty(\mathbb{C})$ $\|1-\bar\partial f\|_{L^\infty}\ge1$

注3 Weyl 引理 $f\in L^1_{loc}(\Omega)$ $\int_\Omega f\partial_{\bar z}h\,dA(z)=0\ \forall h\in C_0^\infty(\Omega)$ 则 $f\in\mathcal{O}(\Omega)$

10.9 Cauchy 变换与 Beurling 变换的收敛性

1. Weyl 引理的证明 $f\in L^1_{loc}(\Omega)$ $\int_\Omega f\,\partial_{\bar z}h\,dA(z)=0\ \forall h\in C_0^\infty(\Omega)$ 则 $f\in\mathcal{O}(\Omega)$

证明: 取 $\rho\in C_0^\infty(\mathbb{C})$ $\rho\ge0$ $\int_{\mathbb{C}}\rho(z)dA(z)=1$ 满足题述条件 $\rho_\varepsilon(z)=\frac{1}{\varepsilon^2}\rho(\frac{z}{\varepsilon})$ $\Omega_\varepsilon=\{z\in\Omega\mid d(z,\partial\Omega)>\varepsilon\}$

对 u 做光滑考虑. $u_\varepsilon=u*\rho_\varepsilon=\int_{\mathbb{C}}u(z-z)\rho_\varepsilon(z)dA(z)\in C^\infty(\Omega_\varepsilon)$

事实: 在 $L^1_{loc}(\Omega)$ 中 $u_\varepsilon\to u$.

$$\int_B|u_\varepsilon(z)-u(z)|dA(z)=\int_B\int_{\mathbb{C}}|u(z-z)-u(z)|\rho_\varepsilon(z)\,dA(z)\,dA(z)$$

$$\le\int_{\mathbb{C}}\rho_\varepsilon(z)\|u(z-z)-u(z)\|_{L^1(\mathbb{C})}dz\le\sup_{|z|\le\varepsilon}\|u(z-z)-u(z)\|_{L^1(\mathbb{C})}\quad\forall\delta>0$$

取 $h\in C_c(\mathbb{C})$ $\|u-h\|_1<\frac{\delta}{3}$ 则 $\|u(z-z)-u(z)\|_{L^1}\le\frac{2}{3}\delta+\|h(z-z)-h(z)\|_{L^1}\to0$.

另一方面 $\bar\partial_z u_\varepsilon=\bar\partial_z\int_{\mathbb{C}}u(\tilde z)\rho_\varepsilon(z-\tilde z)dA(\tilde z)=\int_{\mathbb{C}}u(\tilde z)\,\bar\partial_z\rho_\varepsilon(z-\tilde z)dA(\tilde z)=0$

下提升为内闭一致收敛. 取定 $K\Subset\Omega$ $K_{\varepsilon_0}\Subset\Omega_\varepsilon$ $K_\varepsilon=\{z\in\Omega_\varepsilon\mid d(z,K)<\varepsilon_0\}$.

由 $|u(z)|\le\frac{1}{\pi r^2}\int_{B(z,\varepsilon_0)}|u(z)|\,dA(z)$ $\sup_K|u_\varepsilon-u_\delta|\le\frac{1}{\pi r^2}\|u_\varepsilon-u_\delta\|_{L^1(K_\varepsilon)}$

从而 $\{u_\varepsilon\}$ 内闭一致收敛于某全纯函数 F. 又收敛于 u. u 与 F 差零测集相等.

Date / /

例：单位圆盘示性函数 $\chi_{\mathbb{D}}$ 的 Beurling 变换

解：$S\chi_{\mathbb{D}}(z) = -\frac{1}{\pi}\lim_{\varepsilon\to 0^+}\int_{|z-\zeta|>\varepsilon}\frac{\chi_{\mathbb{D}}(\zeta)}{(z-\zeta)^2}\,dA(\zeta)$

(i). 对 $|z|\geq 1$ 取 $z=Re^{i\phi}$ $\zeta=re^{i\theta}$

$$-\frac{1}{\pi}\int_{|z-\zeta|>\varepsilon}\frac{\chi_{\mathbb{D}}(\zeta)}{(z-\zeta)^2}\,dA(\zeta) = -\frac{1}{\pi}\int_0^{2\pi}\int_0^1 \frac{1}{(Re^{i\phi}-re^{i\theta})^2}\,r\,dr\,d\theta = -\frac{1}{\pi}\int_0^1\int_0^{2\pi}\sum_{n=0}^{\infty}(n+1)\left(\frac{r}{R}e^{i(\theta-\phi)}\right)^n d\theta\,\frac{r\,dr}{(Re^{i\theta})^2}$$

$$= -\frac{1}{\pi}\int_0^1\int_0^{2\pi}\frac{r}{(Re^{i\phi})^2}\,dr\,d\theta = -\frac{1}{z^2}$$

(ii). $|z|<1$. 由 Stokes 公式 $\int_\Omega \partial_{\bar\zeta}F\,dA(\zeta) = -\frac{1}{2i}\int_{\partial\Omega}F\,d\zeta$ (F全纯)

取 $F=\frac{1}{z-\zeta}$ $\partial_{\bar\zeta}F=\frac{1}{(z-\zeta)^2}$

$$-\frac{1}{\pi}\int_{\mathbb{D}\setminus B(z,\varepsilon)}\frac{dA(\zeta)}{(z-\zeta)^2} = \frac{1}{2\pi i}\left(\oint_{|\zeta|=1}\frac{d\bar\zeta}{z-\zeta} - \oint_{|z-\zeta|=\varepsilon}\frac{d\bar\zeta}{z-\zeta}\right) = \frac{1}{2\pi i}\left(\oint_{|\zeta|=1}\frac{d\zeta}{\zeta^2(\zeta-z)} - \oint_{|z-\zeta|=\varepsilon}\frac{(\bar z-\bar\zeta)\,d\bar\zeta}{\varepsilon^2}\right) = 0$$

综上 $S\chi_{\mathbb{D}} = \begin{cases} -\frac{1}{z^2}, & |z|\geq 1 \\ 0, & |z|<1 \end{cases}$

2. ① 引理：$\forall\phi$ 紧可积，$\mathcal{L}\phi := \frac{2}{\pi}\int_{\mathbb{C}}\phi(\zeta)\log|z-\zeta|\,dA(\zeta)$ $\mathcal{C}\phi = \frac{1}{\pi}\int_{\mathbb{C}}\frac{\phi(\zeta)}{z-\zeta}\,dA(\zeta)$ 几乎处处收敛

想法：取 F 紧 证 $\int_F\int_{\mathbb{C}}|\phi(\zeta)\log|z-\zeta||\,dA(\zeta)\,dA(z)<+\infty$ $\int_F\int_{\mathbb{C}}\left|\frac{\phi(\zeta)}{z-\zeta}\right|dA(\zeta)\,dA(z)$ 也 $<+\infty$

($z=\zeta+\varepsilon e^{i\theta}$ 换元 + Fubini 即可)

② 极大变换 $(S_*f)(z) = \frac{1}{\pi}\sup_{\varepsilon>0}\left|\int_{|z-\zeta|>\varepsilon}\frac{f(\zeta)}{(z-\zeta)^2}\,d\zeta\right|$

引理：$\forall f\in L^p(\mathbb{C})$ $1<p<+\infty$，$\int_{|z-\zeta|>\varepsilon}\frac{f(\zeta)}{(z-\zeta)^2}\,d\zeta$ 收敛

证明：$\left|\int_{|z-\zeta|>\varepsilon}\frac{f(\zeta)}{(z-\zeta)^2}\,dA(\zeta)\right| \leq \left(\int_{|z-\zeta|>\varepsilon}|f(\zeta)|^p\,dA(\zeta)\right)^{\frac{1}{p}}\left(\int_{|z-\zeta|>\varepsilon}\left|\frac{1}{(z-\zeta)^2}\right|^{\frac{p}{p-1}}dA(\zeta)\right)^{\frac{p-1}{p}} \leq C r^{-\frac{2}{p-1}}\Big|_{\varepsilon}^{\infty} < +\infty$

3. ① 引理：$\forall f\in L^2(\mathbb{C})$，$Sf\in L^2(\mathbb{C})$

证明：取 $f_n\xrightarrow{L^2} f$ $f_n\in C_0^\infty(\mathbb{C})$ 则 $\|Sf_n - Sf_m\| = \|f_n - f_m\|\to 0$，$Sf_n\to Sf$

② 引理：$\int_{\mathbb{C}}\eta(\zeta)\,S\phi(\zeta)\,dA(\zeta) = \int_{\mathbb{C}}\phi(\zeta)\,S\eta(\zeta)\,dA(\zeta)$ $\forall\eta,\phi\in L^2(\mathbb{C})$

证明：若 $\phi,\eta\in C_0^\infty(\mathbb{C})$ 由对称性，对 $L^2(\mathbb{C})$ 中 η,ϕ 取 $\eta_n\to\eta$ $\phi_n\to\phi$ $\eta_n,\phi_n\in C_0^\infty(\mathbb{C})$

$$\left|\int\eta_n S\phi_n\,dA - \int\eta S\phi\,dA\right| \leq \int|\eta_n-\eta)|S\phi|\,dA + \int|\eta|\,|S(\phi_n-\phi)|\,dA \leq \|\eta_n-\eta\|_{L^2}\|\phi_n\|_{L^2} + \|\eta\|_{L^2}\|\phi_n-\phi\|_{L^2}\to 0$$

③.引理：$-\frac{1}{\pi}\int_{|z-u|>\varepsilon}\frac{f(z)}{(z-u)^2}dA(z)=\frac{1}{|B_\varepsilon(u)|}\int_{B_\varepsilon(u)}Sf$

证明：$S\chi_{B_r}(a)=\begin{cases}0 & |z-a|<r\\ -\frac{r^2}{(z-a)^2}, & |z-a|\geq r\end{cases}$

则 $-\frac{1}{\pi}\int_{|z-u|>\varepsilon}\frac{f(z)}{(z-u)^2}dA(z)=\frac{1}{\pi\varepsilon^2}\int_{\mathbb{C}}f\cdot S\chi_{B_\varepsilon(u)}dA(z)=\frac{1}{\pi\varepsilon^2}\int_{\mathbb{C}}\chi_{B_\varepsilon(u)}Sf\,dA(z)=\frac{1}{|B_\varepsilon(u)|}\int_{B_\varepsilon(u)}Sf$

④.定理：$\forall f\in L^p(\mathbb{C})\ 1<p<\infty\quad (Sf)(z)=-\frac{1}{\pi}\lim_{\varepsilon\to 0}\int_{|z-u|>\varepsilon}\frac{f(z)}{(z-u)^2}dA(z)$ 几乎处处收敛

事实上在 $Sf\in L^p(\mathbb{C})$ 的 Lebesgue 点收敛

4. Lebesque 微分引理定理

① Vitali 覆盖引理：$\mathcal{B}$ 为 $\mathbb{R}^n$ 中一族球 $R=\sup\{\mathrm{rad}(B):B\in\mathcal{B}\}<+\infty$

则存在可数(致)个两两不交的球 $S\subset\mathcal{B}$ 使 $\bigcup_{B\in\mathcal{B}}B\subseteq\bigcup_{B\in S}5B$

② 可测集 E 的 Vitali 覆盖：$\forall x\in E,\delta>0\ \exists B\in V,\ x\in B\ \mathrm{diam}B<\delta$

Vitali 覆盖定理(引)：E，Vitali 覆盖 V 则存在可数(致)个两两不交球 $B_j\in V$ 使 $|E\setminus\bigcup B_j|=0$

③ Lebesque 微分定理

$f\in L^1(\mathbb{R}^d)$ 紧支 $\forall x\in\mathbb{R}^k\quad \tilde{f}(x)=\lim_{r\to 0^+}\frac{1}{\lambda(B(x,r))}\int_{B(x,r)}|f|d\lambda$

则 $\forall\alpha>0,\exists\lambda(\{x\in\mathbb{R}^d|\tilde{f}(x)>\alpha\})\leq\frac{1}{\alpha}\|f\|_{L^1}$

证明：取 $\forall$ 令 $E=\{x\in\mathbb{R}^d|\tilde{f}(x)>\alpha\}$ 取 $*$ 覆盖 $\mathcal{B}$，由闭球 B 满足 $\frac{1}{\lambda(B)}\int_B|f|d\lambda>\alpha$ 组成

取可数集 $\mathcal{C}$ 两两不交 $\lambda(E\setminus\bigcup_{C\in\mathcal{C}}C)=0$.

$$\lambda(E)\leq\sum_{B\in\mathcal{C}}\lambda(B)\leq\sum_{B\in\mathcal{C}}\frac{1}{\alpha}\int_B|f|d\lambda<\frac{1}{\alpha}\int_{\mathbb{C}}|f|d\lambda=\frac{1}{\alpha}\|f\|_{L^1}$$

④.Lebesque 微分定理：$f\in L^1_{loc}(\mathbb{R}^d)$ 则 对几乎处处 $x\in\mathbb{R}^d$

$$f(x)=\lim_{r\to 0^+}\frac{1}{\lambda(B(x,r))}\int_{B(x,r)}f\,d\lambda$$

证明：取 $g_n\in C_c(\mathbb{R}^d)$ $\|f-g_n\|_{L^1}\to 0$ 对 $h\in L^1(\mathbb{R}^d)$ 令

$$Mh(x)=\lim_{r\to 0}\frac{1}{\lambda(B(x,r))}\int_{B(x,r)}|h(y)|\,d\lambda(y)$$

则 $\lim_{r\to 0^+}\left|\frac{1}{\lambda(B(x,r))}\int_{B(x,r)}f(y)d\lambda(y)-f(x)\right|\leq M(f-g_n)(x)+|g_n(x)-f(x)|$

$$\lambda(I_\varepsilon)\leq *\,\lambda\{x:M(f-g_n)(x)\geq\tfrac{\varepsilon}{2}\}+\lambda\{x:|g_n(x)-f(x)|\geq\tfrac{\varepsilon}{2}\}\leq\frac{2}{\varepsilon}\|f-g_n\|_{L^1}+\frac{2}{\varepsilon}\|f-g_n\|_{L^1}\xrightarrow{n\to\infty}0$$

10.16 Fourier变换　　注：查 Margulis 引理

1. ① 速降函数的 Schwarz 类 $\mathcal{S}(\mathbb{C})=\{f\in C^\infty(\mathbb{C}) \mid \lim_{z\to\infty}|z|^m \frac{\partial^{k+l}f(z)}{\partial z^k \partial \bar{z}^l}=0 \ \forall k,l,m\in\mathbb{N}\}$

② 定义 $\mathcal{S}(\mathbb{C})$ 上的 Fourier 变换

$$\mathcal{F}(f)(\zeta)=\hat{f}(\zeta)=\int_{\mathbb{C}} e^{\pi i(z\bar{\zeta}+\bar{z}\zeta)} f(z)\,dA(z)=\int_{\mathbb{R}^2} e^{2\pi i(x\xi+y\eta)} f(x+iy)\,dx\,dy$$

③ 引理(Fourier 逆变换) $f(z)=\int_{\mathbb{C}} e^{-\pi(z\bar{\zeta}+\bar{z}\zeta)}\hat{f}(\zeta)\,dA(\zeta)$　$\hat{\hat{f}}(z)=f(-z)$

证明：由控制收敛定理

$$\int_{\mathbb{C}} e^{-i\pi|\varepsilon|^2(z\bar{\zeta}+\bar{z}\zeta)}\hat{f}(\zeta)\,dA(\zeta)=\lim_{\varepsilon\to 0}\int_{\mathbb{C}} e^{-i\pi(z\bar{\zeta}+\bar{z}\zeta)} e^{-\pi|\zeta|^2\varepsilon^2}\hat{f}(\zeta)\,dA(\zeta)\triangleq\lim_{\varepsilon\to 0} I_\varepsilon(z)$$

$$I_\varepsilon(z)=\int_{\mathbb{C}} e^{-\pi i(z\bar{\zeta}+\bar{z}\zeta)} e^{-\pi\varepsilon^2|\zeta|^2}\int_{\mathbb{C}} f(w) e^{i\pi(w\bar{\zeta}+\bar{w}\zeta)}\,dA(w)\,dA(\zeta)$$

$$=\int_{\mathbb{C}} f(w)\int_{\mathbb{C}} e^{-\pi\varepsilon^2|\zeta|^2-\pi i[(z-w)\bar{\zeta}+\overline{(z-w)}\zeta]}\,dA(\zeta)\,dA(w)$$

（$\int_{\mathbb{R}} e^{-\pi\varepsilon x^2-2\pi i\,\mathrm{Re}(z-w)x}\,dx\int_{\mathbb{R}} e^{-\pi\varepsilon y^2-2\pi i\,\mathrm{Im}(z-w)y}\,dy$）

$$=\int_{\mathbb{C}} f(w)\frac{1}{\sqrt{\varepsilon}} e^{-\frac{\pi}{\varepsilon}\mathrm{Re}^2(z-w)}\frac{1}{\sqrt{\varepsilon}} e^{-\frac{\pi}{\varepsilon}\mathrm{Im}^2(z-w)}\,dA(w)=\int_{\mathbb{C}} f(w) e^{-\pi\frac{|z-w|^2}{\varepsilon}}\frac{dA(w)}{\varepsilon}$$

$$=\int_{\mathbb{C}} e^{-\pi|u|^2} f(z-\sqrt{\varepsilon}u)\,du\to f(z).$$

④ 引理：$\mathcal{F}(f_z)(\zeta)=-i\pi\zeta\hat{f}(\zeta)$　　$\hat{f}_{\bar{\zeta}}(\zeta)=i\pi\,\mathcal{F}(zf)(\zeta)$　推论：$\mathcal{F}:\mathcal{S}\to\mathcal{S}$

$\mathcal{F}(f_{\bar{z}})(\zeta)=-i\pi\bar{\zeta}\hat{f}(\zeta)$　　$\hat{f}_{\zeta}(\zeta)=i\pi\,\mathcal{F}(\bar{z}f)(\zeta)$　　(可将 ∂_z, $\bar{z}$ 移入)

证明：

$$\mathcal{F}(f_{\bar{z}})(\zeta)=\int_{\mathbb{C}} e^{i\pi(z\bar{\zeta}+\bar{z}\zeta)} f_{\bar{z}}(z)\,dA(z)=-\int_{\mathbb{C}}\frac{\partial}{\partial\bar{z}} e^{i\pi(z\bar{\zeta}+\bar{z}\zeta)} f(z)\,dA(\zeta)=-i\pi\zeta\hat{f}(\zeta)$$ 其余同理.

2. $L^1\cap L^2$ 空间上 Fourier 变换的性质

① 定义 ~~$C_0(\hat{\mathbb{C}})$~~ $C_0(\hat{\mathbb{C}})=\{\phi:\hat{\mathbb{C}}\to\mathbb{C}\ \text{连续} \mid \lim_{\zeta\to 0}\phi(\zeta)=0\}$.

② 引理：$\mathcal{F}:L^1(\mathbb{C})\to C_0(\hat{\mathbb{C}})\subset L^\infty(\mathbb{C})$

证明：$|\hat{f}(z)|=\left|\int_{\mathbb{C}} e^{i\pi(z\bar{\zeta}+\bar{z}\zeta)} f(z)\,dA(\zeta)\right|\le\int_{\mathbb{C}}|H(z)|\,dA(z)=\|f\|_{L^1}\Rightarrow\|\hat{f}\|_{L^\infty}\le\|f\|_{L^1}$

Campus

问：为何存在？

$\forall \varepsilon>0$ 由 $f\in L^1$，取 $g\in \mathcal{S}(\mathbb{C})$ $\|f-g\|_{L^1}<\frac{1}{2}\varepsilon$ 则 $\|\hat f-\hat g\|_{L^\infty}\le \|f-g\|_{L^1}<\frac{1}{2}\varepsilon$

而 $\lim_{|z|\to 0}\hat g(z)=0$ $|\hat f(z)|\le |\hat f(z)-\hat g(z)|+|\hat g(z)|<\varepsilon$ $|z|\to+\infty$.

③ 引理：$C_0(\hat{\mathbb{C}})$ 在逐点乘法下构成代数。与 L^1 在卷积下构成代数在 $\mathcal{F}$ 下为同构

· $\|\phi\psi\|_{L^\infty}\le \|\phi\|_{L^\infty}\cdot\|\psi\|_{L^\infty}$

· $f*g(z)=\int_{\mathbb{C}} f(z-w)g(w)\,dA(w)$ $\widehat{(f*g)}(z)=\hat f(z)\hat g(z)$

$\|f*g\|_{L^1}\le \|f\|_{L^1}\cdot\|g\|_{L^1}$

证明：（原证明用逼近 $f_n\to f$ $g_n\to g$ 但似不必要）

$$\widehat{f*g}(z)=\int_{\mathbb{C}}\int_{\mathbb{C}} f(z-w)g(w)\,dA(w)\,e^{i\pi(z\bar z+\bar z z)}\,dA(z)$$

$$=\int_{\mathbb{C}} e^{i\pi(w\bar z+\bar w z)}g(w)\,dA(w)\int_{\mathbb{C}} f(z-w)e^{i\pi[(z-w)\bar z+(\bar z-\bar w)z]}\,dA(z)=\hat g(z)\cdot\hat f(z)$$

④. 引理（Parseval 等式） $\langle\hat\phi(z),\hat f(z)\rangle=\langle\phi,f\rangle$

证明：$\langle\hat\phi,\hat{\bar f}\rangle=\int_{\mathbb{C}}\phi(z)\int_{\mathbb{C}}\overline{f(z)}e^{-\pi i(z\bar z+\bar z z)}\,dA(z)\,dA(z)=$

$$=\int_{\mathbb{C}}\int_{\mathbb{C}}\hat\phi(z)e^{-\pi i(z\bar z+\bar z z)}\,dA(z)\,\overline{f(z)}\,dA(z)=\int_{\mathbb{C}}\hat\phi\bar f\,dA(z)=\langle\phi,f\rangle.$$

推论：$\mathcal{F}$ 为 $L^2(\mathbb{C})$ 上的酉算子 $\|\mathcal{F}f\|_{L^2}=\|f\|_{L^2}$

⑤. 注记：用 "interpolation" 方法，可以证明 $\mathcal{F}$ 为 L^p 到 L^q 连续算子 $1\le p\le 2\le q\le\infty$，$\frac{1}{p}+\frac{1}{q}=1$

3. 乘积算子与 Hecke 恒等式

①. $m=m(z)$ 为有界可测函数 定义 $T_m: L^2(\mathbb{C})\to L^2(\mathbb{C})$ $T_m f=\mathcal{F}^{-1}(m\hat f)$

满 $\|T_m f\|_{L^2}=\|m\hat f\|_{L^2}\le\|m\|_{L^\infty}\cdot\|f\|_{L^2}$

注：通常只考虑 m 满足 $m(\lambda z)=m(z)$ $\lambda\in\mathbb{R}^+$ 且 m 在 0 处光滑

注（Marcinkiewicz 定理）可将定义域延拓至 L^p 空间

②. 令 $P_k(z)=P_k(x,y)$ 为复值齐次调和多项式，k 次 $(az^k+b\bar z^k)$

取 $m=\frac{2\pi i^k}{k}\frac{P_k(z)}{|z|^k}$ ~~则 T_m 称 Hecke 算子~~.

KOKUYO

③. 引理: $T_m\phi(z)=\lim_{\varepsilon\to 0}\int_{|z-w|>\varepsilon}\frac{P_k(z-w)}{|z-w|^{k+2}}\phi(w)\,dw$.

证明: 用 $\phi\in C_0^\infty(\mathbb{C})$ 逼近 $L^2(\mathbb{C})$

Step1 先证右式存在. 令 $K(w)=\frac{P_k(w)}{|w|^{k+2}}$ $K(re^{i\theta})=\frac{P_k(e^{i\theta})}{r^2}$ 由调和 $\int_0^{2\pi}P(e^{i\theta})d\theta=0$

从而 $\int_{\varepsilon<|w|<R}K(w)\,dw=0$

$$RHS=\int_{|w|>\varepsilon}K(w)\phi(z-w)\,dw=\int_{\varepsilon<|w|<1}K(w)(\phi(z-w)-\phi(z))\,dA(w)+\int_{|w|>1}K(w)\phi(z-w)\,dA(w)$$

(I_1: $\int_{\varepsilon<|w|<1}$, I_2: $\int_{|w|>1}$)

$$|I_1|\le\int_{\varepsilon<|w|<1}|K(w)|\cdot C|w|\,dA(w)\le C\cdot\int_0^1\frac{1}{r^2}\cdot r\cdot r\,dr<+\infty$$

$$|I_2|<C\int_1^\infty\frac{1}{r^2}r\,dr<+\infty$$

Step2: 用~~光滑~~ Gauss 分布逼近. 令 $\frac{1}{|w|^{k+2}}=A_k\int_0^\infty t^{\frac{k}{2}}e^{-\pi t|w|^2}dA\,dt$ $A_k=\frac{\pi^{1+\frac{k}{2}}}{\Gamma(1+\frac{k}{2})}$

$RHS=\int_{\mathbb{C}}K(w)[\phi(z-w)-\phi(z)\chi(w)]\,dA(w)$ 其中 $\chi\in C_0^\infty(\mathbb{C})$ 在0的邻域1 $\chi(re^{i\theta})=\chi(r)$

Fubini $A_k\int_0^\infty t^{\frac{k}{2}}dt\int_{\mathbb{C}}P_k(w)e^{-\pi t|w|^2}[\phi(z-w)-\phi(z)\chi(w)]\,dA(w)$

$=A_k\int_0^\infty t^{\frac{k}{2}}(g_t*\phi)(z)\,dt$ $g_t=P_k(w)e^{-\pi t|w|^2}$

断言: $\hat{g}_t(\zeta)=i^k t^{-k-1}P_k(\zeta)e^{-\frac{\pi|\zeta|^2}{t}}$

理由 由 $P_k(z)=az^k+b\bar{z}^k$ 只需证 $P_k=z^k$ 对 $z=u+iv$

$$\hat{g}_t(\zeta)=\mathcal{F}(z^ke^{-\pi t|z|^2})=\int_{\mathbb{R}^2}(x+iy)^ke^{-\pi t(x^2+y^2)+2\pi i(xu+yv)}dx\,dy$$

$$=\int_{\mathbb{R}^2}(x-\tfrac{iu}{t}+\tfrac{iv}{t}+i\tfrac{u+iv}{t})^ke^{-\pi t[(x+\frac{iu}{t})^2+(y+\frac{iv}{t})^2]-\frac{\pi}{t}(u^2+v^2)}dx\,dy$$

$$=e^{-\frac{\pi|\zeta|^2}{t}}\int_{-\infty-\frac{iv}{t}}^{+\infty-\frac{iv}{t}}\int_{-\infty-\frac{iu}{t}}^{-\infty-\frac{iu}{t}}(x+iy+\tfrac{i\zeta}{t})^ke^{-\pi t(x^2+y^2)}dx\,dy$$

$$=e^{-\frac{\pi|\zeta|^2}{t}}\int_{\mathbb{R}^2}\sum_{j=0}^k(x+iy)^j\left(\tfrac{i\zeta}{t}\right)^{k-j}\binom{k}{j}e^{-\pi t(x^2+y^2)}dx\,dy$$

$$=e^{-\frac{\pi|\zeta|^2}{t}}\int_{\mathbb{R}}\left(\tfrac{i\zeta}{t}\right)^ke^{-\pi t(x^2+y^2)}dx\,dy=i^kt^{-k-1}\zeta^ke^{-\pi\frac{|\zeta|^2}{t}}$$

代入 $RHS=A_k\int_0^\infty t^{\frac{k}{2}}\int_{\mathbb{C}}e^{-i\pi(z\bar{\zeta}+\bar{z}\zeta)}\hat{g}_t(\zeta)\hat{\phi}(\zeta)\,dA(\zeta)\,dt$

$$=A_k\int_0^\infty i^kt^{-\frac{k}{2}-1}\int_{\mathbb{C}}e^{-i\pi(z\bar{\zeta}+\bar{z}\zeta)}P_k(\zeta)e^{-\pi\frac{|\zeta|^2}{t}}\hat{\phi}(\zeta)\,dA(\zeta)\,dt$$

Campus

$= \int_{\mathbb{C}} e^{-i\pi(z\bar{\zeta}+\bar{z}\zeta)} \frac{2\pi i^k P_k(\zeta)}{k} \frac{}{|\zeta|^k} \hat{\phi}(\zeta)\, dA(\zeta) \quad = \mathcal{F}^{-1}(m\hat{\phi}) = T_m\phi(z)$

注: $A_k \int_0^\infty t^{-\frac{k}{2}-1} e^{-\pi\frac{|y|^2}{t}} dt \xrightarrow{s=\frac{t}{\pi}} -\frac{\pi^{1+\frac{k}{2}}}{\Gamma(1+\frac{k}{2})} \int_\infty^0 s^{\frac{k}{2}-1} e^{-\pi|y|^2 s} ds = * \frac{\pi^{1+\frac{k}{2}}}{\Gamma(1+\frac{k}{2})} \frac{1}{\pi^{\frac{k}{2}}} \int_0^\infty u^{\frac{k}{2}-1} e^{-u} du$

$= * \frac{\Gamma(\frac{k}{2})\pi}{\Gamma(1+\frac{k}{2})} = \frac{2\pi}{k}.$

注: $\widehat{\left(\frac{P_k(z)}{|z|^{k+2}}\right)} = \frac{2\pi i^k P_k(z)}{|z|^k}$ $\mathcal{F}\left(\frac{P_k(z)}{|z|^{k+2}}\right) \mathcal{F}(\phi(z))$

(想法 $\mathcal{F}\left(\frac{P_k(z)}{|z|^{k+2}} * \phi(z)\right) = \frac{2\pi i^k}{k} \frac{P_k(y)}{|y|^k} \mathcal{F}(\phi)$)

例: 对 $P_k(z) = \bar{z}^2$ $m \Leftrightarrow \frac{\bar{z}^2}{|z|^4} = \frac{z^2}{|z|^2}$ 即 Beurling 算子.

例: $\widehat{\left(\frac{z^k}{|z|^{k+2}}\right)} = \frac{z^k}{|z|^k}$

④ 定义奇异积分算子 $R^k: L^p \to L^p$ $(R^k\phi)(z) = \frac{k}{2\pi i^k} \lim_{\varepsilon\to 0} \int_{|z-y|>\varepsilon} \frac{(\bar{z}-\bar{\zeta})^k}{|z-y|^{k+2}} \phi(y)\, dy$

定理: Beurling 变换 S $S\phi(z) = -\frac{1}{\pi} \lim_{\varepsilon\to 0} \int_{|z-y|>\varepsilon} \frac{\phi}{(z-\zeta)^2} dA(\zeta)$ 为 $m_2 = \frac{z^2}{|z|^2}$ 的乘积算子

⑤ 引理: m 的积算子伴随为 $\bar{m}$ 积算子 (L^2 中) $R_k^* = R_{-k}$

证明:

$$\langle T_m f, g\rangle = \langle \widehat{T_m f}, \hat{g}\rangle = \int_{\mathbb{C}} m(\zeta)\hat{f}(\zeta)\overline{\hat{g}(\zeta)}\, dA(\zeta) = \int_{\mathbb{C}} \hat{f}(\zeta)\overline{\bar{m}(\zeta)\hat{g}(\zeta)}\, dA(\zeta)$$

$$= \langle \hat{f}, \widehat{\bar{m}g}\rangle = \langle f, T_m g\rangle \Rightarrow T_m^* = T_m \Rightarrow R_k^* = R_{-k}.$$

八声甘州 柳永

对潇潇暮雨洒江天,一番洗清秋。渐霜风凄紧,关河冷落,残照当楼。是处红衰翠减,苒苒物华休。惟有长江水,无语东流。

不忍登高临远,望故乡渺邈,归思难收。叹年来踪迹,何事苦淹留。想佳人,妆楼颙望,误几回,天际识归舟。争知我,倚栏杆处,正恁凝愁!

10.23 Beurling 变换的 L^p-理论

1. ① 定义 $BMO(\Omega)$ 为 Ω 上局部可积函数 满足

$$\|u\|_* = \sup_{B\subseteq\Omega} \frac{1}{|B|} \int_B |u-u_B|\, dA(z) < \infty$$

其中 B 为 Ω 中闭圆盘, $u_B = \frac{1}{|B|}\int_B u\, dA(z)$.

注. BMO 范数为半范数, 模常数 (构成 Banach 空间)

KOKUYO

② 引理：$\phi \in L^1(\mathbb{C})$ 则 $\|\mathcal{L}\phi\|_* \le \frac{2}{\pi}\|\log|\cdot|\|_* \cdot \|\phi\|_{L^1}$ 即 $\mathcal{L}: L^1(\mathbb{C}) \to BMO(\mathbb{C})$

证明：用 $\phi \in C_0^\infty$ 逼近

$$(\mathcal{L}\phi)_B = \frac{1}{|B|}\int_B \mathcal{L}\phi(z)\,dA(z) = \frac{2}{\pi}\int_{\mathbb{C}} \frac{1}{|B|}\int_B \log|z-\zeta|\,\phi(\zeta)\,dA(\zeta)\,dA(z) = \frac{2}{\pi}\int_{\mathbb{C}} (\log|\cdot|)_{B-\zeta}\phi(\zeta)\,dA(\zeta)$$

$$\mathcal{L}\phi(z) - (\mathcal{L}\phi)_B = \frac{2}{\pi}\int_{\mathbb{C}} [\log|z-\zeta| - (\log|\cdot|)_{B-\zeta}]\phi(\zeta)\,dA(\zeta)$$

$$\frac{1}{|B|}\int_B |\mathcal{L}\phi(z) - (\mathcal{L}\phi)_B|\,dA(z) \le \frac{2}{\pi}\int_{\mathbb{C}} |\phi(\zeta)| \frac{1}{|B-\zeta|}\int_{B-\zeta} |\log|w| - (\log|B\cdot|)_{B-\zeta}|\,dA(w)\,dA(\zeta)$$

$$\le \frac{2}{\pi}\int_{\mathbb{C}} |\phi(\zeta)|\,\|\log|\cdot|\|_*\,d\zeta = \frac{2}{\pi}\|\phi\|_{L^1}\cdot\|\log|\cdot|\|_*.$$

对 $\phi_j \in C_0^\infty(\mathbb{C})$ $\phi_j \xrightarrow{L^1} \phi$ $\|\mathcal{L}\phi_j - \mathcal{L}\phi_k\|_* \le \frac{2}{\pi} C\|\phi_j - \phi_k\|_{L^1} \to 0$. 从而 $\mathcal{L}\phi = \lim_{j\to\infty} \mathcal{L}\phi_j$

③ 性质 (i). $L^\infty(\mathbb{C}) \subsetneq BMO(\mathbb{C})$ $\left(\frac{1}{|B|}\int |f - f_B|\,dA \le 2\|f\|_{L^\infty} < +\infty\right)$

(ii). $BMO(\mathbb{C}) \not\subset L^p(\mathbb{C})$ 如 $|$ $L^p(\mathbb{C}) \not\subset BMO(\mathbb{C})$

(iii) $BMO(\mathbb{C}) \subset \bigcap_{1\le p<\infty} L^p_{loc}(\mathbb{C})$ ← John Nirenberg 不等式

④ 引理. $\log|z| \in BMO(\mathbb{C})$

证明：对 $B = B(a,r)$ 按 a, r 大小关系分类讨论

· $|a| \ge 2r$ $\forall z \in B$ $\frac{1}{2} \le \frac{|z|}{|a|} \le \frac{3}{2}$ $\Rightarrow |\log|z| - \log|a|| \le \log 2$

$$\int_B \log|z|\,dA(z) = \int_0^{2\pi}\int_0^r t\log|a+te^{i\theta}|\,dt\,d\theta = \int_0^r\int_0^{2\pi} t\log t + t\log\left|\frac{a}{t}+e^{i\theta}\right|\,d\theta\,dt$$

$$= \int_0^r 2\pi t\log t + 2\pi t\log\left|\frac{a}{t}\right|\,dt = \int_0^r 2\pi t\log|a|\,dt = \pi r^2\log|a|$$

故 $(\log|\cdot|)_B = \log|a|$ $\frac{1}{|B|}\int_B |\log|z| - (\log|\cdot|)_B|\,dA \le \log 2$.

· $r < |a| < 2r$. $B = B(a,r) \subset B(0,3r)$

$$\frac{1}{|B|}\int_B \log|z|\,dA(z) = \frac{1}{\pi r^2}\int_0^r\int_0^{2\pi} t\log|a+te^{i\theta}|\,d\theta\,dt = \log|a|.$$

$$\frac{1}{|B|}\int_B |\log|z| - \log|a||\,dA(z) \le \frac{1}{\pi r^2}\int_B |\log|z| - \log 3r|\,dA + \frac{1}{\pi r^2}\int_B |\log 3r - \log|a||\,dA$$

$$\le \frac{1}{\pi r^2}\int_{|z|<3r} \log\frac{3r}{|z|}\,dA(z) + \frac{1}{\pi r^2}\int_B \log 3\,dA = \frac{9}{2} + \log 3$$

· $|a| < r$. $\frac{1}{|B|}\int_B \log|z|\,dA(z) = \frac{1}{\pi r^2}\int_0^r\int_0^{2\pi} t\log|a+te^{i\theta}|\,d\theta\,dt = \frac{1}{\pi r^2}\int_{|a|}^r 2\pi t\log|a|\,dt = \frac{|a|^2}{r^2}\log|a|$

注：$\frac{1}{|B|}\int_B |u - u_B|\,dA \le \frac{2}{|B|}\int_B |u-c|\,dA$ $\forall c$. ($\because |u-u_B| \le |u-c| + |c-u_B| \le |u-c| + \frac{1}{|B|}\int_B |u-c|\,dA$).

~~$\frac{1}{|B|}\int_B |\log|z| - \frac{|a|^2}{r^2}\log|a|\,|\,dA(z) \le \frac{1}{\pi r^2}\int_B |\log|z| - \log 3r|\,dA + \frac{1}{\pi r^2}\int_B |\log 3r - \frac{|a|^2}{r^2}\log|a|\,|\,dA$~~

$= \frac{1}{\pi r^2}\int_0^{|a|} 2\pi t\log|a|\,dt + \frac{1}{\pi r^2}\int_{|a|}^{r} 2\pi t\log t\,dt$. $\left(\int t\log t\,dt = \frac{t^2}{2}\log t - \frac{t^2}{4} + C\right)$

$= \frac{|a|^2}{r^2}\log|a| + \frac{2}{r^2}\left(\frac{r^2}{2}\log r - \frac{r^2}{4} - \frac{|a|^2}{2}\log|a| + \frac{|a|^2}{4}\right) = \log r - \frac{1}{2} + \frac{|a|^2}{2r^2}$

$\frac{1}{|B|}\int_B |\log|z| - (\log)_B|\,dA \le \frac{1}{|B|}\int_B |\log|z| - \log 3r|\,dA + \frac{1}{|B|}\int_B |\frac{1}{2} - \frac{|a|^2}{2r^2} - \log 3|\,dA$.

$\le \frac{9}{2} + \frac{1}{2} + \log 3 = 5 + \log 3$ 有界. 综上 $\log|z| \in BMO(\mathbb{C})$

⑤ 引理: $h(z) = \begin{cases} (\log|z|)^2, & 0<|z|<1 \\ 0, & |z|>1 \end{cases} \in L^1(\mathbb{C})$ $h \notin BMO(\mathbb{C})$

证明: 取 $B_r = B(0,r)$ ~~由计~~ $h_{B_r} = \frac{1}{\pi r^2}\int_0^r\int_0^\theta d\theta\cdot(\log\rho)^2\rho\,d\rho = (\log r)^2 - \log r + \frac{1}{2}$.

在 $B_{\frac{r}{e}}$ 上 $|h(z) - h_{B_r}| \ge |(\log r - 1)^2 - ((\log r)^2 - \log r + \frac{1}{2})| \ge \log\frac{1}{r} + \frac{1}{2}$. ~~$r\to 0$时~~

$\frac{1}{|B|}\int_B |h(z) - h_{B_r}| \ge \frac{1}{\pi r^2}\cdot 2\pi\int_0^{\frac{r}{e}} \log\frac{1}{\rho} + \frac{1}{2}\,d\rho \to +\infty$ $r\to 0$ 故 $h \notin BMO(\mathbb{C})$.

例: 多项式均不在 $BMO(\mathbb{C})$ 中 (调和)

⑥ 引理: $\mathrm{Re}\,z \notin BMO(\mathbb{C})$ (调和. B_R 上平均值 0)

证明: $\frac{1}{|B_R|}\int_{B_R} |\mathrm{Re}\,z|\,dA(z) = \frac{1}{\pi r^2}\int_0^{2\pi} d\theta\int_0^R r|\cos\theta|\,r\,d\theta = \frac{R}{2\pi}\int_0^{2\pi}|\cos\theta|\,d\theta \to +\infty$.

⑦ 引理 (i). $u \in BMO(\mathbb{C})$. $F: \mathbb{C}\to\mathbb{C}$ $|F(z) - F(y)| \le L|z-y|$ 则 $F\circ u \in BMO(\mathbb{C})$

(ii). $u \in BMO(\mathbb{C})$ 实值 F Lip 连续 则 $F\circ u \in BMO(\mathbb{C})$ (取 $F(u+iv) = F(u)$ 即可)

(iii). $\sqrt{\log(e+|z|^2)} \in BMO(\mathbb{C})$ (取 $u = \log|z|$ $F(t) = \sqrt{\log(e+e^{2t})}$ 即可.

证明 (i). $|F(u)| \le F(0) + L|u|$. $u \in BMO(\mathbb{C})$ $F\circ u \in L^1_{loc}$

$\|F\circ u\|_* = \frac{1}{|B|}\int_B |F(u) - (F\circ u)_B|\,dA \le \frac{1}{|B|}\int_B |F(u) - F(u_B)|\,dA + \frac{1}{|B|}\int_B |F(u_B) - (F\circ u)_B|\,dA$

$= \frac{1}{|B|}\int_B |F\circ u(z) - F(u_B)|\,dA + |F(u_B) - \frac{1}{|B|}\int_B F\circ u(z)\,dA(z)|$

$\le \frac{2}{|B|}\int_B |F(u) - F(u_B)|\,dA \le \frac{2L}{|B|}\int_B |u - u_B|\,dA \le 2L\cdot\|u\|_*$.

KOKUYO

例: $\chi_{\mathbb{D}}=\begin{cases}1, & z\in\mathbb{D}\\ 0, & z\notin\mathbb{D}\end{cases}$ 的对数势

解: $\mathcal{L}\chi_{\mathbb{D}}(z)=\frac{2}{\pi}\int_{|\zeta|<1}\log|z-\zeta|\,dA(\zeta)=\frac{2}{\pi}\int_0^1\rho\int_0^{2\pi}\log|re^{i\theta}-\rho e^{i\theta}|\,d\theta\,d\rho$

$=\frac{2}{\pi}\int_0^1\rho\cdot 2\pi\log\max\{|z|,\rho\}\,d\rho=\begin{cases}2\log|z|, & |z|\geq 1\\ |z|^2-1, & |z|<1\end{cases}$

对一般 $B(a,r)$ 类似有 $\mathcal{L}\chi_{B(a,r)}(z)=\begin{cases}2r^2\log|z-a|, & z\notin B(a,r)\\ |z-a|^2+2r^2\log r-r^2, & z\in B(a,r)\end{cases}$

引理

⑧命题: $\mathcal{L}\chi_{\mathbb{D}}\in BMO(\mathbb{C})$ 且 $\mathcal{L}\chi_{\mathbb{D}}\notin VMO(\mathbb{C})$

证明: 取 $B=B(0,R)$ $R\to\infty$. $f=\mathcal{L}\chi_{\mathbb{D}}$

$$f_{B(0,R)}=\frac{1}{\pi R^2}\left(\int_0^1(r^2-1)r\,dr\cdot 2\pi+\int_1^R 2\log r\cdot r\,dr\cdot 2\pi\right)=2\log R-1+\frac{1}{2R^2}$$

$$\frac{1}{|B(0,R)|}\int_{B(0,R)}|f-f_{B(0,R)}|=\frac{1}{\pi R^2}\left[\int_0^1\left|r^2-2\log R+1-\frac{1}{2R^2}\right|r\,dr\cdot 2\pi+\int_1^R\left|2\log r-2\log R+1-\frac{1}{2R^2}\right|r\,dr\,2\pi\right]$$

有界. 且 $\geq\frac{\pi}{4}>0$. $R\to+\infty$.

例: $\chi_{\mathbb{D}}$ 的 Cauchy 变换

解: $\mathcal{C}\chi_{\mathbb{D}}=\frac{1}{\pi}\int_{|\zeta|<1}\frac{1}{z-\zeta}\,d\zeta$

· $|z|>1$. $\mathcal{C}\chi_{\mathbb{D}}(z)=\frac{1}{\pi}\sum_{n=0}^{\infty}\frac{1}{z^{n+1}}\int_{|\zeta|\leq 1}\zeta^n\,dA(\zeta)=\frac{1}{z}$

· $|z|\in(0,1)$ $\mathcal{C}\chi_{\mathbb{D}}(z)=\frac{1}{\pi}\int_0^1 r\,dr\int_0^{2\pi}\frac{d\theta}{z-re^{i\theta}}=\pi\bar{z}$

$$\mathcal{C}\chi_{\mathbb{D}}(z)=\begin{cases}\frac{1}{z}, & |z|>1\\ \bar{z}, & 0\leq|z|<1\end{cases}$$

对一般 $B(a,r)$ 类似有 $\mathcal{C}\chi_{B(a,r)}=\begin{cases}\frac{r^2}{z-a}, & |z-a|>r\\ \overline{z-a}, & z\in B(a,r).\end{cases}$

2. ①. 定义: $VMO(\Omega)$ 为 $BMO(\mathbb{C})$ 中 $C_0^\infty(\mathbb{C})$ 的闭包.

$$VMO=\left\{f:\mathbb{C}\to\mathbb{C}\;\middle|\;\lim_{r\to 0}\sup_{\mathrm{rad}(B)<r}\frac{1}{|B|}\int_B|f-f_B|=0\text{ 且 }\lim_{R\to\infty}\sup_{\mathrm{rad}(B)>R}\int_B|f-f_B|=0\right\}.$$

②. 定理: $\mathcal{C}:L^2(\mathbb{C})\to VMO(\mathbb{C})$. $\|\mathcal{C}\phi\|_*\leq 2\|\phi\|_{L^2}$.

证明: 先证明一个事实 $\mathcal{C}\phi=\frac{1}{\pi}\int_{\mathbb{C}}\left[\frac{1}{z-\zeta}+\frac{\chi_{\mathbb{C}\setminus\mathbb{D}}}{z}\right]\phi(\zeta)\,dA(\zeta)$ 几乎处处收敛.
(用于处理∞)

固定 $R>\max\{|z|,1\}$. 对 $|z|>2R$

$$\int_{|\zeta|>2R}\left(\frac{1}{z-\zeta}+\frac{1}{z}\right)\phi(\zeta)\,dA(\zeta)\leq\|\phi\|_{L^2}\left(\int_{|\zeta|>2R}\left|\frac{z}{(z-\zeta)\zeta}\right|^2dA(\zeta)\right)^{\frac{1}{2}}\leq C_{onst}\|\phi\|_{L^2}\text{ 有界}$$

而 $\int_{\mathbb{C}} 1_{1\le|z|\le 2R} \frac{\phi(z)}{|z|} dA(z) \le \int_{|z|<2R} |\phi(z)| dA(z) \le \|\phi\|_{L^2} < +\infty$

从而只需估计 $\int_{|z|<2R} \frac{\phi(z)}{|z-z|} dA(z)$. 几乎处处收敛. 对 z 积分

$\int_{|z|<R}\int_{|z|<2R} \frac{\phi(z)}{|z-z|} dA(z)dA(z) \le \int_{|z|<2R} |\phi(z)| dA(z) \cdot \int_{|w|<3R} \frac{dA(w)}{|w|} < +\infty$. 故几乎处处收敛

问题：若 $C_\phi f_j \to X \in BM O(G)$ $X \overset{?}{=} C_\phi$. (可以再引进一串算子逼近)

需 $\frac{1}{\pi}\int_{\mathbb{D}} |C_\phi(z) - (C_\phi)_{\mathbb{D}}| dA(z) \le 2\|\phi\|_{L^2}$

$(C\phi)_{\mathbb{D}} = \frac{1}{\pi}\int_{\mathbb{C}} \phi(z) \frac{1}{\pi}\int_{\mathbb{D}} \frac{dA(z)}{z-z} dA(z) = \frac{1}{\pi}\int_{\mathbb{C}} \phi(z) \overline{C\chi_{\mathbb{D}}(z)} dA(z)$

$= -\frac{1}{\pi}\int_{\mathbb{D}} \bar{z}\phi(z) dA(z) - \frac{1}{\pi}\int_{\mathbb{C}\setminus\mathbb{D}} \frac{\phi(z)}{z} dA(z)$.

$C\phi(z) - (C\phi)_{\mathbb{D}} = \frac{1}{\pi}\int_{\mathbb{D}} (\frac{1}{z-z} + \bar{z})\phi(z) dA(z) + \frac{1}{\pi}\int_{\mathbb{C}\setminus\mathbb{D}} |\frac{1}{z-z} + \frac{1}{z}| \phi(z) dA(z)$.

$LHS \le \frac{1}{\pi}\int_{\mathbb{C}} |\phi(z)||\psi(z)| dA(z) \le \|\phi\|_{L^2} \cdot \|\psi\|_{L^2}$.

其中 $\psi(z) = \begin{cases} \frac{1}{\pi}\int_{\mathbb{D}} |\frac{1}{z-z} + \bar{z}| dA(z) \le |z| + 4. & |z| \le 1 \\ \frac{1}{\pi}\int_{\mathbb{D}} |\frac{1}{z-z} + \frac{1}{z}| dA(z) \le \frac{1}{\pi|z|}\int_{\mathbb{D}} \frac{|z|}{|z-z|} dA(z) \le 4. & 1 \le |z| \le 2 \\ \le \frac{2}{|z|^2}. & |z| \ge 2 \end{cases}$ L^2 收敛.

注：$\cdot \frac{1}{\pi}\int_{\mathbb{D}} |\frac{1}{z-z} + \bar{z}| dA(z) \le \frac{1}{\pi}\int_{B(z,2)} \frac{1}{|z-z|} dA(z) + \frac{1}{\pi}\int_{\mathbb{D}} |\bar{z}| dA$

$\le \frac{1}{\pi} \cdot 2\pi \int_0^2 \frac{1}{r} r dr + \frac{1}{\pi} \cdot \pi |z| = |z| + 4$ $|z| < 1$

$\cdot \frac{1}{\pi|z|}\int_{\mathbb{D}} \frac{|z|}{|z-z|} dA \le \frac{1}{\pi|z|}\int_{\mathbb{D}} \frac{dA}{|z-z|} \le \frac{1}{\pi}\int_{B(z,2|z|)} \frac{dA}{|z-z|} = \frac{1}{\pi} \cdot 2\pi \int_0^{2|z|} \frac{1}{r} r dr = 4$.

$\cdot \frac{1}{\pi|z|}\int_{\mathbb{D}} \frac{|z|}{|z-z|} dA \le \frac{1}{\pi|z|}\int_{\mathbb{D}} \frac{2}{|z|} dA = \frac{2}{|z|^2}$ 因 $|\frac{z}{z} - 1| \ge 1 - \frac{|z|}{|z|} \ge \frac{1}{2} \ge \frac{|z|}{2}$ 故 $\frac{|z|}{|z-z|} \le \frac{2}{|z|}$.

10.28. Cauchy 算子的 L^p 理论

1. 定理：$\phi \in L^2(\mathbb{C})$ 则 $\exists$ 弱导数 $(C\phi)_z = \phi$ $(C\phi)_{\bar{z}} = S\phi$

(注：弱导数 $\langle \frac{\partial}{\partial z^k}\frac{\partial}{\partial \bar{z}^l} f, h\rangle = (-1)^{k+l}\int_{\mathbb{C}} f \frac{\partial}{\partial z^k}\frac{\partial}{\partial \bar{z}^l} h\, dA(z)$)

证明：对 $\psi \in C_0^\infty(\mathbb{C})$

$\|\mathcal{C}\phi\|_{L^2} = \|\phi\|_{L^2}$

$\int_{\mathbb{C}} \psi (\mathcal{C}\phi)_z = -\int_{\mathbb{C}} \psi_z (\mathcal{C}\phi) \overset{*}{=} \int_{\mathbb{C}} \mathcal{C}(\psi_z)\phi \overset{\text{对换}}{=\!=} \int_{\mathbb{C}} (\mathcal{C}\psi)_z \phi = \int_{\mathbb{C}} \mathcal{S}\psi\, \phi \overset{\downarrow}{=} \int_{\mathbb{C}} \psi\, \mathcal{S}\phi$

下证 (*) 成立. 由 $\int_{\mathbb{C}} \psi_z = 0$. 由 3(iii) $\mathcal{C}\psi_z \in L^2(\mathbb{C})$. 取一列 $\phi_k \in C_0^\infty$ $\phi_k \xrightarrow{L^2} \phi$

$\int_{\mathbb{C}} \psi_z \mathcal{C}\phi_k \, dA(z) = \frac{1}{\pi} \int_{\mathbb{C}} \int_{\mathbb{C}} \frac{\psi_z \phi_k(\zeta)}{z-\zeta} dA(\zeta) dA(z) = -\int_{\mathbb{C}} \phi_k(\zeta) \mathcal{C}\psi_z \, dA(\zeta)$.

而 $\left| \int \phi_k \mathcal{C}\psi_z - \int \phi \mathcal{C}\psi_z \right| \le \|\phi_k - \phi\|_{L^2} \cdot \|\mathcal{C}\psi_z\|_{L^2} \to 0$. 从而 $\int_{\hat{\mathbb{C}}} \psi_z (\mathcal{C}\phi) = -\int_{\mathbb{C}} \phi (\mathcal{C}\psi_z)$

2. 定理: 对 $1 < q < 2 < p < +\infty$. $\frac{1}{p} + \frac{1}{q} = 1$ 则 Cauchy 变换为 $L^p \cap L^q \to C_0(\hat{\mathbb{C}})$ 的连续线性映射

其中 $C_0(\hat{\mathbb{C}}) = \{\phi : \hat{\mathbb{C}} \to \mathbb{C} \mid \phi$ 连续. $\lim_{z \to \infty} \phi(z) = 0\}$.

且 $\|\mathcal{C}\phi\|_{L^\infty} \le \frac{2}{\sqrt{2-q}} \sqrt{\|\phi\|_p \|\phi\|_q} \le \frac{1}{\sqrt{2-q}} (\|\phi\|_p + \|\phi\|_q)$

证明: $|\mathcal{C}\phi(z)| \le \left| \frac{1}{\pi} \int_{|\zeta - z| \le R} \frac{\phi(\zeta)}{z-\zeta} dA(\zeta) \right| + \left| \frac{1}{\pi} \int_{|\zeta - z| \le R} \frac{\phi(\zeta)}{z-\zeta} dA(\zeta) \right|$

$\le \frac{1}{\pi} \|\phi\|_p \left(\int_{|\zeta - z| \le R} \frac{d\zeta}{|\zeta - z|^q} \right)^{\frac{1}{q}} + \frac{1}{\pi} \|\phi\|_q \left(\int_{|\zeta - z| \ge R} \frac{d\zeta}{|z-\zeta|^p} \right)^{\frac{1}{p}}$.

注: $\|f\|_{L^2}^2 \le \|f\|_p \|f\|_q$.

3. 定理: $1 < p < +\infty$. $\phi \in L^p(\mathbb{C})$. $\phi(\zeta) = 0$ $\forall |\zeta| \ge R$. 则

(i). $\|\mathcal{C}\phi\|_{L^p(B(0,2R))} \le 6R \|\phi\|_p$ (ii) $\left\| \mathcal{C}\phi - \frac{1}{\pi z} \int \phi \right\|_{L^p(\mathbb{C} \setminus D_{2R})} \le \frac{2R}{(p-1)^{\frac{1}{p}}} \|\phi\|_p$

特别 (iii). 若 $\int_{\mathbb{C}} \phi = 0$. 则 $\|\mathcal{C}\phi\|_{L^p(\mathbb{C})} \le \frac{8R}{(p-1)^{\frac{1}{p}}} \|\phi\|_p$

(iv). 若 $p > 2$. 则 $\|\mathcal{C}\phi\|_{L^p(\mathbb{C})} \le (6 + 3(p-2)^{-\frac{1}{p}}) R \|\phi\|_p$

证明: (i). 对 $z \in B(0,2R)$. $(\mathcal{C}\phi)_z = \frac{1}{\pi} \int_{B(0,R)} \frac{\phi(\zeta)}{z-\zeta} dA(\zeta) = \int_{\mathbb{C}} \frac{\chi_{B(0,3R)}(z-\zeta)}{\pi(z-\zeta)} \phi(\zeta) dA(\zeta)$

由 Young 不等式 $\|\mathcal{C}\phi\|_{L^p(B(0,2R))} \le \left\| \frac{\chi_{B(0,3R)}(z-\zeta)}{\pi(z-\zeta)} \right\|_{L^1} \|\phi\|_{L^p} = 6R \|\phi\|_{L^p}$

(ii). 对 $|z| > 2R$. $|\zeta| < R$ $\left| \mathcal{C}\phi(z) - \frac{1}{\pi z} \int \phi \right| = \frac{1}{\pi} \left| \int_{\mathbb{C}} \left(\frac{1}{z-\zeta} - \frac{1}{z} \right) \phi(\zeta) dA(\zeta) \right| \le \frac{2R}{\pi |z|^2} \|\phi\|_{L^1} \le \frac{2R (\pi R^2)^{\frac{1}{p}}}{\pi |z|^2} \|\phi\|_p$

$\left\| \mathcal{C}\phi - \frac{1}{\pi z} \int \phi \right\|_{L^p(\mathbb{C} \setminus D_{2R})} \le \frac{2R}{(p-1)^{\frac{1}{p}}} \|\phi\|_p$

(iii). 直接求和　　(iv). $\|\mathcal{C}\phi\|_{L^p(\mathbb{C}\setminus B(0,2R))} \le \frac{2}{(p-2)^{\frac{1}{p}}} R\|\phi\|_p$ ← $p>2$时不减 $\int$也可积

注(Young不等式) $\frac{1}{p}+\frac{1}{q}=1+\frac{1}{r}$　$f\in L^p$　$g\in L^q$　则

$$\|f*g\|_{L^r} \le \|f\|_{L^p}\cdot\|g\|_{L^q}$$

证明: 取$\alpha_1=\frac{1}{r}$　$\alpha_2=\frac{1}{p}-\frac{1}{r}$　$\alpha_3=\frac{1}{p}-\frac{1}{r}$　由Hölder不等式

$$\left|\int_{\mathbb{R}^n} f(y)g(x-y)dy\right| \le \left(\int_{\mathbb{R}^n}|f(y)|^p|g(x-y)|^q dy\right)^{\alpha_1}\left(\int_{\mathbb{R}^n}|f(y)|^p dy\right)^{\alpha_2}\left(\int_{\mathbb{R}^n}|g(x-y)|^q dy\right)^{\alpha_3}$$

$$\Rightarrow \int_{\mathbb{R}^n}\left|\int_{\mathbb{R}^n} f(y)g(x-y)dy\right|^r dx \le \int_{\mathbb{R}^n}\int_{\mathbb{R}^n}|f(y)|^p|g(x-y)|^q dx dy\ \|f\|_{L^p}^{r-p}\|g\|_{L^q}^{r-q}$$

$$\Rightarrow \|f*g\|_{L^r}^r \le \|f\|_{L^p}^p\cdot\|g\|_{L^q}^q\cdot\|f\|_{L^p}^{r-p}\|g\|_{L^q}^{r-q} = \left(\|f\|_{L^p}\|g\|_{L^q}\right)^r$$

4. ① 定义Hölder空间 $C^\alpha(\mathbb{C})$　$0<\alpha\le 1$.

$$C^\alpha(\mathbb{C}) = \{\eta:\mathbb{C}\to\mathbb{C}\text{连续} \mid \|\eta\|_{C^\alpha(\mathbb{C})} = \sup_{z\neq w}\frac{|\eta(z)-\eta(w)|}{|z-w|^\alpha}<\infty\}$$ ← 可逐点估计

性质 · $C^\alpha(\mathbb{C})$ Banach空间 (模常数意义下)

· $C^\infty(\mathbb{C})$ 在$C^\alpha(\mathbb{C})$不稠密　如$|z|^\alpha \notin \overline{C^\infty(\mathbb{C})}$

· $\forall\phi\in C^\infty(\mathbb{C})$　$\exists\phi_k\in C^\infty(\mathbb{C})$　$\|\phi_k\|_{L^\alpha}<C$且 $\lim_{k\to\infty}\|\phi_k-\phi\|_{C^\infty}=0$

(卷积 $\varphi_k(w)=k^2\varphi(kw)$　$\operatorname{supp}\varphi\subset B(0,1)$ $\int_{\mathbb{C}}\varphi=1$. 令$\phi_k=\varphi_k*\phi$)

问题: 哪些函数可被C^∞函数C^α逼近?

② 定理: $2<p<\infty$. 则 Cauchy变换将$L^p(\mathbb{C})$连续映入C^α　$\alpha=1-\frac{2}{p}$

$$\|\mathcal{C}\phi\|_{C^\alpha(\mathbb{C})} \le \frac{12p^2}{p-2}\|\phi\|_{L^p(\mathbb{C})}$$

证明: 对$z\neq w$

$$\frac{|\mathcal{C}\phi(z)-\mathcal{C}\phi(w)|}{|z-w|^{1-\frac{2}{p}}} = \frac{1}{\pi}\frac{1}{|z-w|^{1-\frac{2}{p}}}\left|\int_{\mathbb{C}}\phi(\zeta)\left(\frac{1}{z-\zeta}-\frac{1}{w-\zeta}\right)dA(\zeta)\right| \le \|\phi\|_{L^p}\cdot\frac{1}{\pi}\left|\int_{\mathbb{C}}\frac{\phi(\zeta)dA(\zeta)}{(z-w)(w-\zeta)}\right|\cdot\frac{z-w}{|z-w|^{1-\frac{2}{p}}}$$

$$\le \frac{1}{\pi}\|\phi\|_{L^p}\left|\int_{\mathbb{C}}\frac{|z-w|^{\frac{p}{p-1}}dA(\zeta)}{|z-\zeta|^{\frac{p}{p-1}}|w-\zeta|^{\frac{p}{p-1}}}\right|^{\frac{p-1}{p}}$$

$$\overset{\zeta=(w-z)\eta}{=\!=} \frac{1}{\pi}\|\phi\|_{L^p}\left|\int_{\mathbb{C}}\frac{dA(\eta)}{|\eta|^{\frac{p}{p-1}}|1-\eta|^{\frac{p}{p-1}}}\right|^{\frac{p-1}{p}} \le \frac{12p^2}{\pi(p-1)}\|\phi\|_p.$$

注: 此处恰对在0.1.∞三个奇点均可奇积

5. 共形几何

① 记 $S(2)$ 为行列式为1的正定对称矩阵 $\left\{\begin{pmatrix} a & b \\ b & c \end{pmatrix} \middle| \overset{a>0}{\downarrow} \text{正定}, ac-b^2=1\right\}$

② Ω 上 Riemann 度量 $A: \Omega \to \mathbb{R}_+ \cdot S(2) \subset \mathbb{R}^3$

$$A(z)=\begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad ds=\langle dz, A(z)dz\rangle^{\frac{1}{2}}=\sqrt{adx^2+2bdxdy+cdy^2}$$

③ 两个 Riemann 结构称共形 若存在 $\phi: \Omega \to \mathbb{R}_+$ 使 $\langle,\rangle_A=\langle,\rangle_B \phi(z)$ 即 $A(z)=B(z)\phi(z)$

Campus